VOLIS

Bringing **order to chaos**

Data Mining II

Volis case study

FACULDADE DE ECONOMIA UNIVERSIDADE DO PORTO

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With you today

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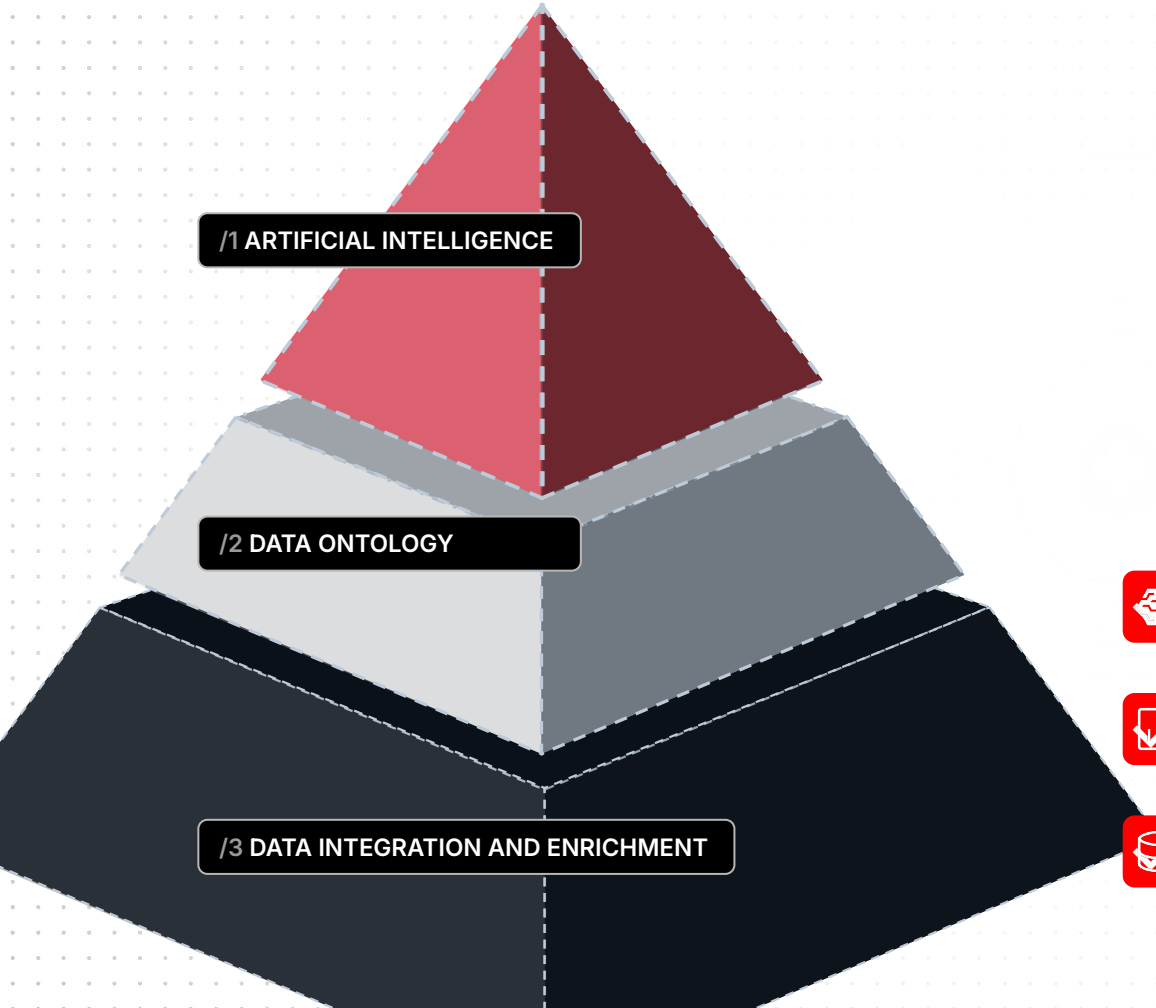
Background: Energy, Bank (Risk), Construction; Health and Industry



OUR APPROACH

Volis Method

It is a structured approach to integrating AI into business operations. It begins with the integration and enrichment of operational data, moves through defining the business ontology and applying advanced analytics on that foundation, and culminates in the development of AI applications integrated into real workflows, supporting critical decisions with scale and security.

- Artificial intelligence
- Data ontology
- Data integration and enrichment

OUR PARTNERS

Some of our partners throughout these past 2 years

We partner with companies across diverse industries

MOTA-ENGIL

SILVER LAKE

Resolute

ia.

AMORIM

CASAIS

SUMA

IDC CLINIC

WHERE WE OPERATE

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Global projects

Volis develops highly complex projects with a global reach. From our operations in four countries, we work with multinational clients whose implementations span more than 20 geographies, ensuring scalable, robust solutions adapted to different operational realities.

20+

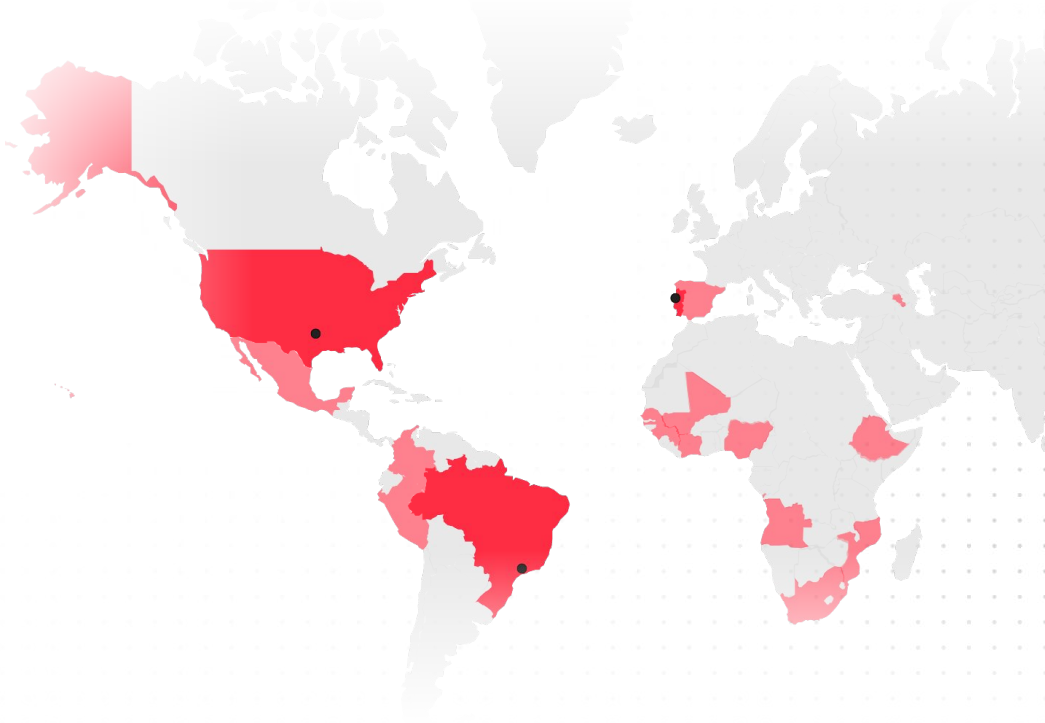
COUNTRIES OF OPERATION

35+

TEAM

3 Porto (HQ), São Paulo (BR), Madrid (SP)

COUNTRIES



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Technology partners that amplify the impact of our solutions

Partners

Palantir

aws

CLOUDFLARE

GitHub

Google

Certifications

STARTUPORTUGAL

IAPMEI
Parcerias para o Crescimento

Tech Stack

Redis
Cadvisor
Airbyte
Portainer
Kubernetes
Node Exporter
Docker
Nginx
Oauth2
Helm
Grafana
Prometheus
Certbot
Pyspark
Terraform
DBT Cloud
Big Query
Jupyter
Metabase
Argo CD
Airflow
Looker
Firebase Analytics
Clojure

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Your challenge

Optimizing Production for AgroMill Corp

CASE CONTEXT

Your team has been hired to optimize operations for your new client AgroMill Corp.

Context

AgroMill Corp production and profitability are highly sensitive to external conditions and raw material characteristics, such as:

Particle Size (Granulometry)
Affects texture, dissolution time, and flavor (e.g., coffee powder or flour)

Density
Crucial for packaging operations to ensure packets are filled accurately by weight.

Moisture Content
The primary factor in preventing mold and ensuring a viable shelf-life.

Goal

Assess which operational components and raw material features most affect AgroMill's production output, providing them with a data-driven strategy to increase efficiency, quality, and throughput

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AGROMILL CORP

It's time to demonstrate your skills from the Data Mining II course.

Be thoughtful, creative, and show why you're the best fit for the task!

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How AgroMill works

Shifts

Morning (06:00 - 14:00)

Afternoon (14:00 - 22:00)

Night (22:00 - 06:00)

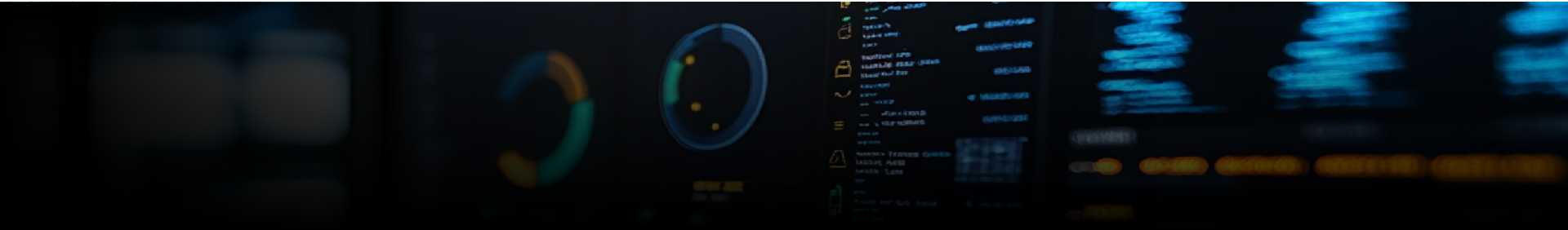
Note: At the end of each shift, a "reset" and a set-up period are required.

Maintenance

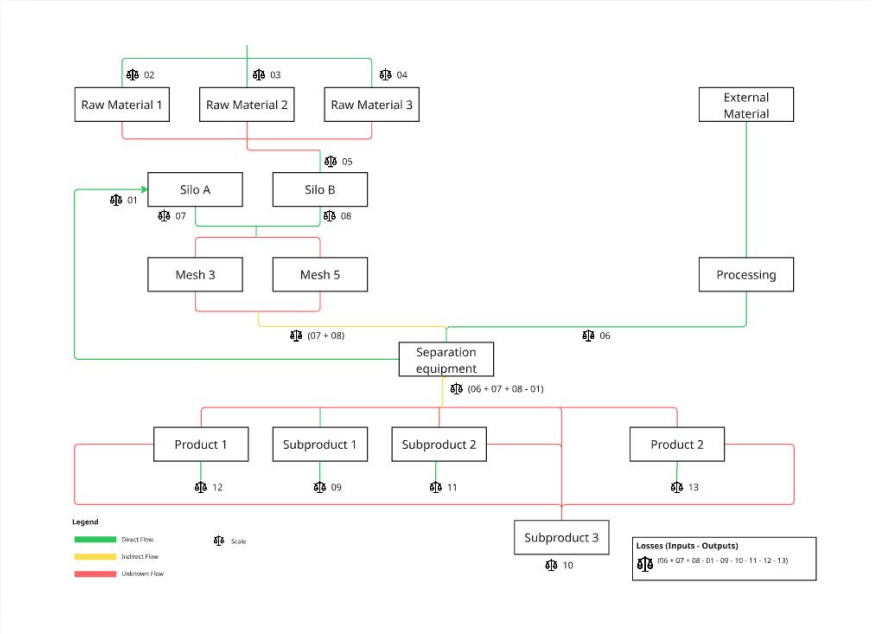
Maintenance and replacement of specific mechanical components, such as the sieves (crivos), can increase set-up time during specific shifts

Quality control

Quality controls are performed continuously during each shift to ensure the product meets AgroMill's client specifications and contractual requirements



Illustrative diagram for the client's processes



Understanding of the client’s processes

Scales

The scales are the **key measurement points** across the production line, recording how much material enters and leaves as final product. They validate the mass balance, detect process deviations, and ensure each fraction is routed correctly. Reliable scale readings are essential for tracking flow, confirming separation performance, and supporting quality control.

Production Variables

Mechanical Drivers: Maintenance of Sieves and equipment separation settings are the primary drivers of output volume

Input/Output Balance: Critical monitoring to ensure physical laws are met and hardware data collection is accurate.

Shift Operations: Three daily shifts (Morning, Afternoon, Night) with mandatory "reset" and setup periods

Sensitivity: High vulnerability to external conditions due to the use of natural raw materials.

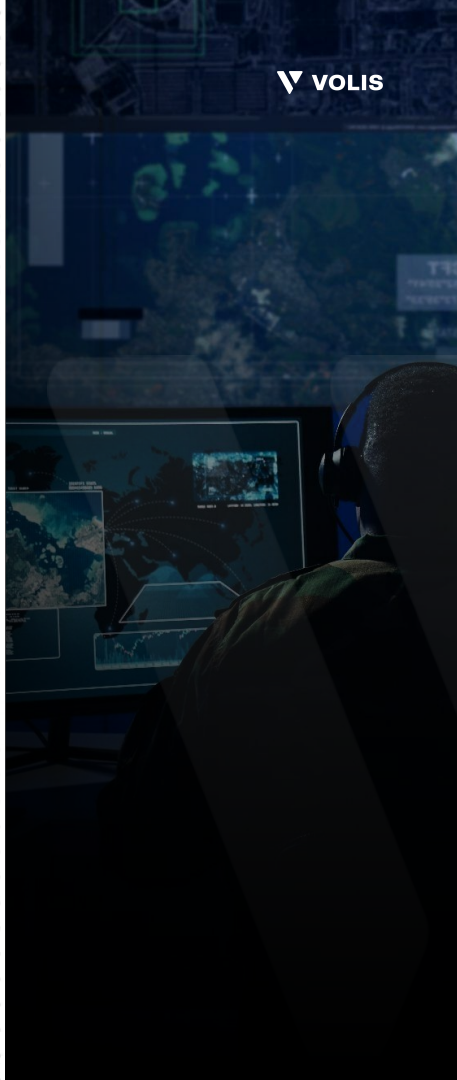
Product Dataset (Output Quality)

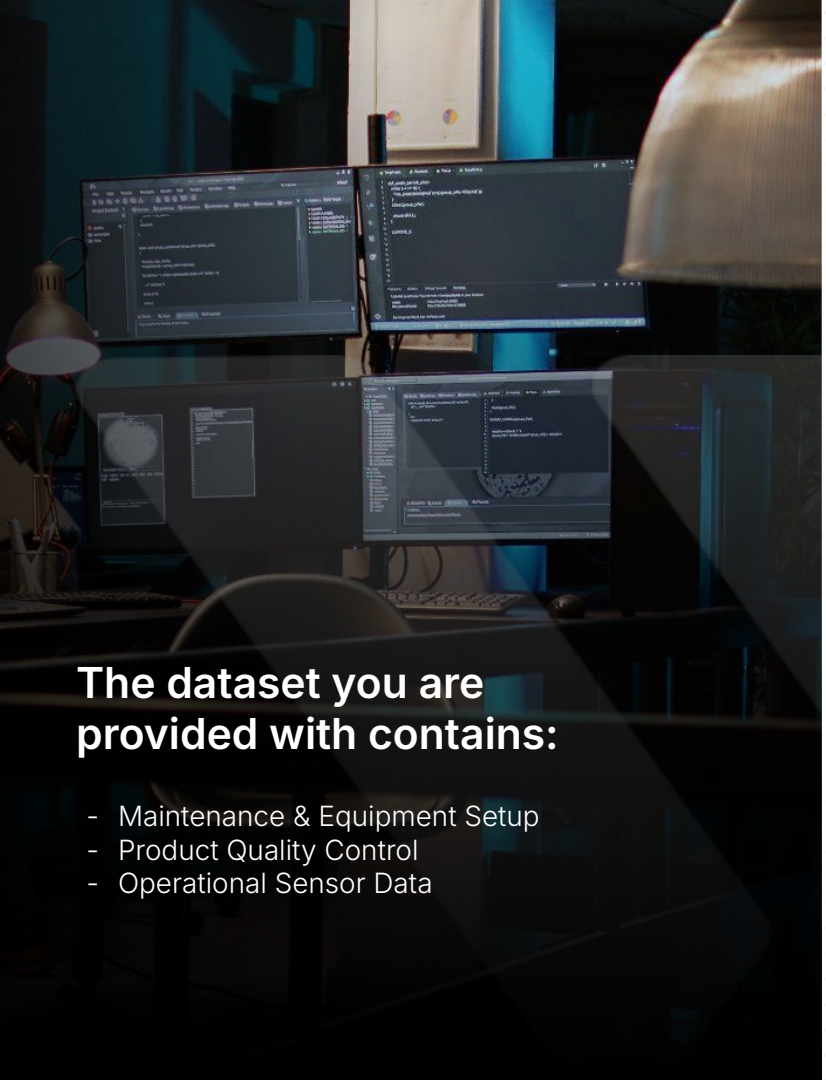
Moisture: Essential for shelf-life and mold prevention.

Particle Size: Determines final texture, flavor, and dissolution.

Density: Controls packaging accuracy and weight-based filling.

Quality Score: Continuous analysis focusing on Product 1 specifications





The dataset you are provided with contains:

- Maintenance & Equipment Setup
- Product Quality Control
- Operational Sensor Data

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A. Maintenance & Equipment Setup

Historical records of mechanical component replacements and "reset" periods across shifts

2 Datasets:

- Equipment separation (5 columns)
- Mesh (8 columns)

B. Product Quality Control

Detailed sample analysis used to determine the Quality Score

3 Datasets:

- Product 1, for two different clients (6 columns)
- Product 2 (6 columns)

C. Operational Sensor Data

Real-time monitoring of factory components and their performance

1 Datasets:

- Sensors (3 columns)

Maintenance, Equipment Setup and Sensors

equipment_separation	mesh	sensor_parquet
id_unidade_prod	data	Tag
id_equipamento_sep	tamanho_calibre_malha_3mm	Value
id_malha_ref	tamanho_calibre_malha_5mm	Date time
espec_abertura_mal	freq_calibre_malha_3mm	
dt_manut_componente	freq_calibre_malha_5mm	
	chefia	

Product Quality Control

Produto_01_2	Product_01	Product_02
data_teste	data_teste	data_teste
id_ensaio	id_ensaio	id_ensaio
Detail	Detail	Detail
data_prod	data_prod	data_prod
densidade	densidade	densidade
humidade	humidade	humidade
origem	origem	origem
8_236, 25_071, 18_100, 14_140, 10_200, Under_00	8_236, 25_071, 18_100, 14_140, 10_200, Under_000	8_236, 25_071, 18_100, 14_140, 10_200, Under_000

*__Produto_01_2__ and __Product_01__ share the same structure and refer to the same product, but they contain **records from different clients. Product_02** also has the same structure, but the measurements correspond to **Product 2**.

Use case guidelines

Feel free to add other analysis if you see fit

A. Data Preparation

Examine the dataset's structure for completeness and consistency and ensure it is ready to analyze, by either cleaning or engineering data

B. Exploratory Data Analysis

Use descriptive statistics to summarize key variables and visualize distributions for better understanding

C. Machine Learning Algorithms

Use a variety of machine learning algorithms to tackle the a series of defined challenges, and provide solution / recommendations

D. Open Reflection

Use relevant skills and research for this topic, applying creativity and logic in your analysis. Develop a clear thesis and support it with data

Deep dive: Data Cleaning

Objective

To begin your assessment you should process the data, ensuring it is ready for any further analyses

Hence, your first challenge is to apply the data cleansing techniques you learnt during the Data Mining course

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Hint: Pay attention to timestamps and physical laws

Proposed methodology

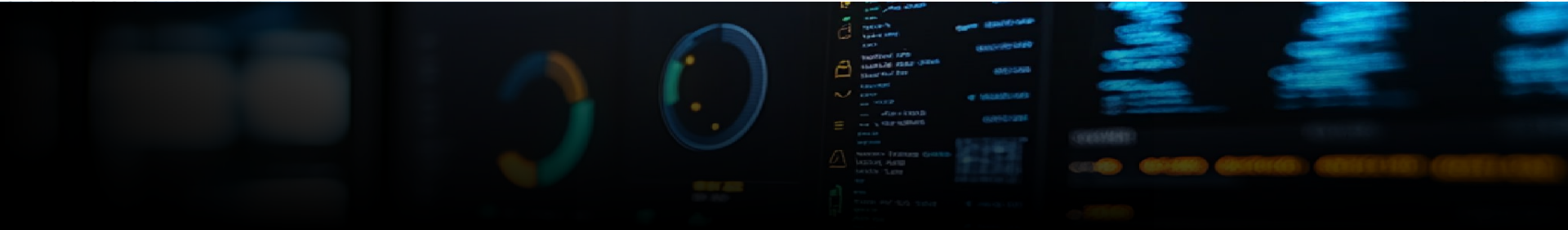
Analyze the dataset to identify and address key data quality issues:

- **Missing Data:** Assess the extent and patterns of incomplete records.
- **Illogical/Impossible Data:** Detect and resolve values that violate physical reality or domain constraints.
- **Potential Anomalies:** Flag and investigate extreme outliers that may distort analysis.

Clearly articulate the rationale and logic for every data cleaning method applied to ensure transparency and justify the cleansed dataset's integrity.

Deep dive: Exploratory Data Analysis

Proposed requirement	**Example of questions to be answered** (not extensive)
Insights	What takeaways can be drawn from descriptive analytics?
Visual Analysis	Are there any underlying concerns or physical bottlenecks presented by the data?
Potential Concerns	Provide visual evidence (eg., correlation matrices, distributions) to support your claims?



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Machine Learning algorithms

A. Quality Score Prediction

Develop a classification or regression model to predict the Quality Score of "Product 1" based on sample analysis.

B. Throughput Estimation

Build regression models to estimate the production volume of the 5 main products and sub-products.

C. Feature Importance & Drivers

Identify the most critical variables driving each of the outcomes.

For all requests explain your model outputs in the most logical way possible

Goals of this challenge

Ability to form and defend a thesis

Create a clear, evidence-based strategy supported by both factory data and external market factors.

Critical thinking and data interpretation

Analyze and interpret industrial patterns to draw relevant conclusions about what truly drives production efficiency.

Challenge intuition vs. data

Question the existing data collection methods (like sensor glitches) if they don't align with physical reality, actively seeking to prove or disprove operational assumptions.

Communication and justification

Assess the ability to express complex ideas in simple, understandable arguments that are easy to follow and debate

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Open Reflection

Based on your modeling, what actions will Volis recommend AgroMill Corp take to extract **more financial value** from their operations?

Furthermore, what should be the **next steps** for your data science team, and what is the long-term vision for advanced analytics in this factory?

All these characteristics will be challenged daily in your future work

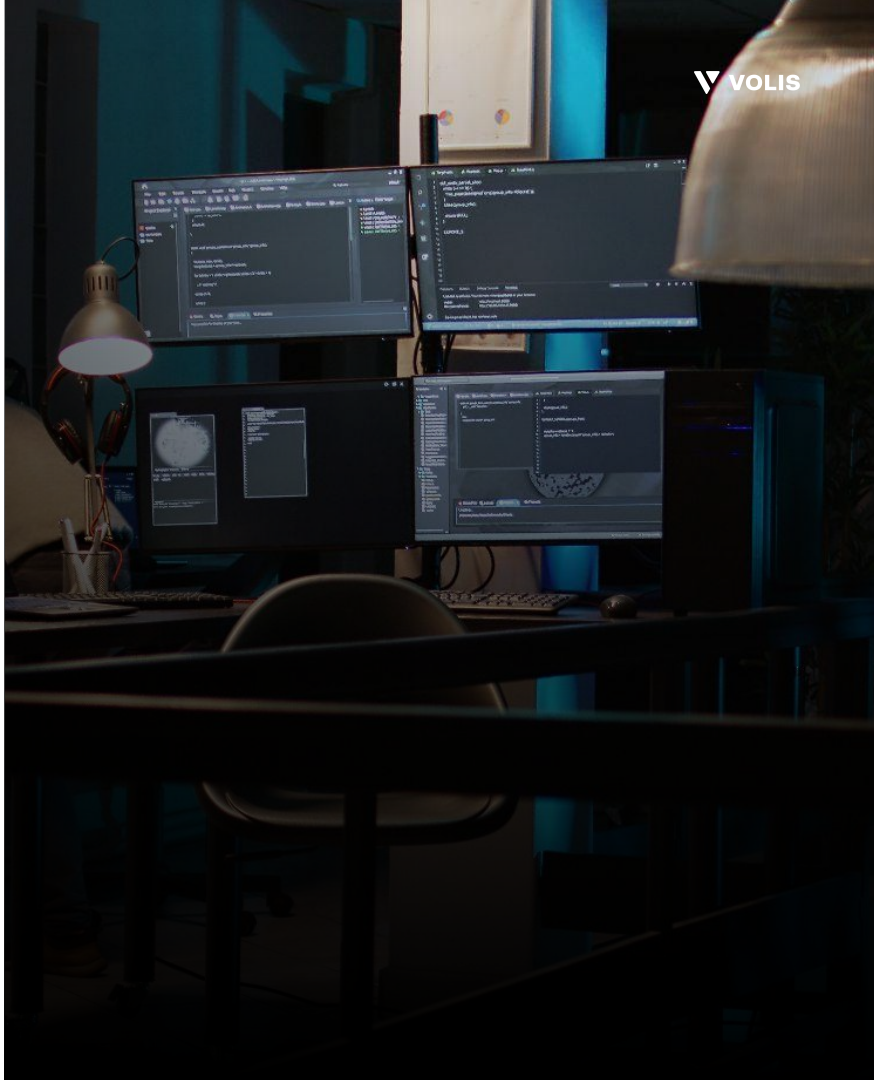
Overview and key details

- Your groups should have the size of 5 elements
- The results and proposed solutions should be presented in 7 weeks
- The presentation should take around 25 minutes and you should expect a 10 minutes period for follow-up questions
- Midway through your journey you will have a session with the Volis team who can guide you and help solve some blockers you may find
- If any questions arise during the process that require immediate attention, you may contact via email: pedro@volis.ai or andre@volis.ai

Thank you
Any questions?

One last note:

- Although it may not always be the main focus of the evaluation, remember to take pride in your presentation materials
- Whether consciously or subconsciously, they impact how your work is perceived
- Invest time to ensure your presentations are distinct, clear, direct, and organized
- This skill translates to the job market



VOLIS

Bringing order to chaos.

We build AI solutions that become part of your business's operating system, from data to decision, in real time

volis.ai